



Fig. 4: PolyCycl plant that claims to achieve 95% efficiency

and more than eighty percent of the polymer hydrocarbons become sustainable energy products. The output hydrocarbon oil is further fractionated into petrol, diesel, and jet fuel that can run petrol and diesel engines. This oil also finds many industrial applications such as the generation of steam, hot water, hot air, or for heating thermic fluid. The oil and the gas can be used in internal combustion engines to produce electricity.

Rudra Environmental Solutions has bagged the Dynamic Asia Global Award for its outstanding contribution to the recycling industry. Rudra uses the process called Thermo Catalytic Depolymerisation (TCD) in which shredded plastics are heated at 380°C to 430°C in absence of oxygen. A catalyst is used to crack the polymer chain, thus producing hydrocarbon vapours, which are condensed. The condensed vapours produce a fuel called polyfuel, and synthetic gases called syngas. The polyfuel undergoes filtration before collection, and the syngas is also cleaned before using in a reactor.

At Rudra, each tonne of plastics produces 600 to 650 litres of polyfuel, which is used in boilers, furnaces, etc. Efficiency of syngas, which is used for heating the reactor at Rudra, is 20%

to 25%. In this process, no harmful gases or effluents are generated. The end product yields a residual char, which is a non-hazardous waste that can be used as a road filler along with bitumen.

The advantage of the method followed at Rudra is that plastic wastes need not be presorted, cleaned or dried before processing. This reduces the overall cost of operation, the process is eco-friendly, and the final output yield is not only energy-efficient but also has a negative carbon footprint.

PolyCycl Private Limited has won the Indian Innovation Initiative award from the Confederation of Indian Industry (CII) for making an impact towards sustainable circular economy by converting waste plastics to high-value petroleum fuels. PolyCycl has also been chosen as one among the eleven winners of the Asia-Pacific End Plastic Waste Innovation Platform, from Alliance to End Plastic Waste & Plug and Play, for innovative ideas from startups in efficiently making use of the plastic wastes.

Amit Tandon, founder of PolyCycl, strongly believes that producing high-value fuel and adopting low-cost extraction technology are the two main factors to increase the

plastic-to-fuel conversion. The plant at PolyCycl adopts Contiflo Cracker extraction technology that uses a patented continuous process to generate distillate fuels, which can be used in industrial boilers and furnaces. In the process of converting waste plastics to liquid fuels and gas, PolyCycl achieves an efficiency of 95%. For every tonne of waste plastics, PolyCycl plant produces 5.8 to 6.4 barrels of liquid fuels. In short, for every ton of plastic wastes the yield is 850 litres of lighter distillate fuels.

According to PolyCycl, extracting fuel from waste plastics using a batch-based approach is not only tedious but gives an output that is of low-value; whereas the fuels generated from PolyCycl's process are lighter, less viscous and high in **GCV (global value chain?)**, thus making them highly effective and valuable. These fuels require no further distillation and are ready to be sold as a substitute for industrial diesel.

Paterson Energy provides a wholesome solution to plastic wastes through a sustainable and eco-friendly mechanism. Paterson uses continuous Thermo Chemical Depolymerisation (TCD) in its recycling plant to break the large plastic molecules, which yields a fuel named by them as 'verte-x' that is free from sulphur-like toxic elements. Verte in French means 'green,' and as the fuel finds many industrial applications, the company has added an 'x' to it.

Paterson produces 600 litres of verte-x from one tonne of plastic wastes. The carbon black residue is used as a replacement of coal. Verte-x can be used as a fuel in generators to produce electricity, in marine engines, in casting furnace, or any other plant that requires heat or electricity.

MK Aromatics, a manufacturer of various aromatic hydrocarbon solvents, also provides a technical solution to synthesise waste plastics or multi-layered plastics like sachets. The company uses a completely automated method where there is no need to segregate or wash or dry the plastics. The plant converts seven to ten tonnes of collected waste plastics per day.

At MK Aromatics, the collected waste plastics are loaded

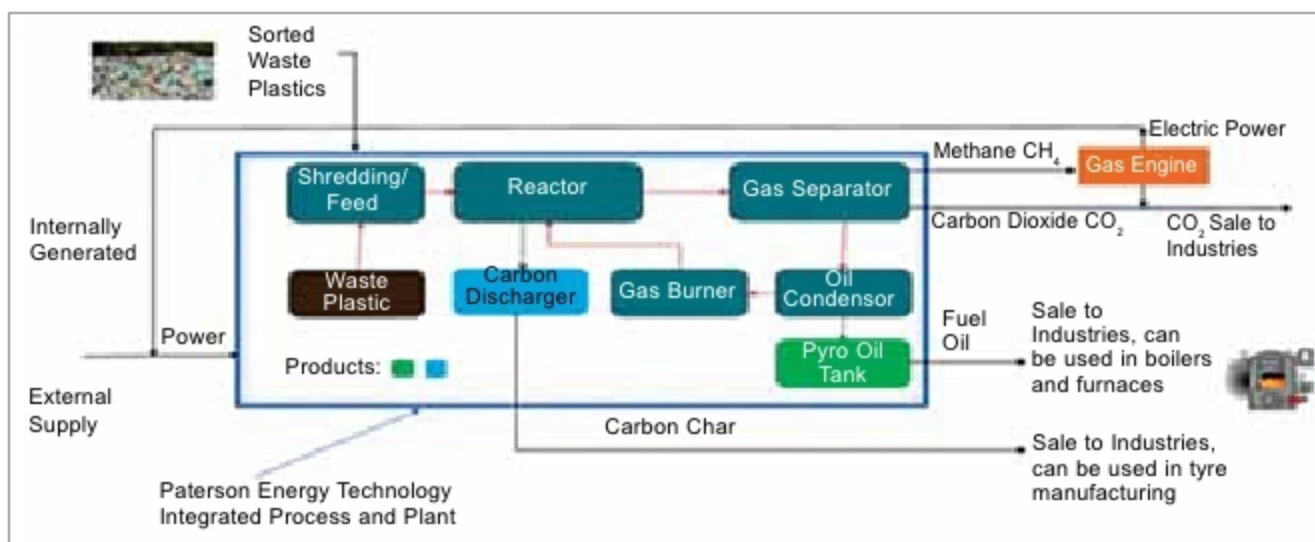


Fig. 5: Plastic waste menace converted to usable products (Credit: Paterson Energy)